HAROUT (HARRY) ALTUNYAN

Co-Founder & CTO · AI Systems Architect · Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Technology founder and full-stack engineer who takes products from concept to production. Co-Founder and CTO of SAECULA, an AI health intelligence platform where I lead architecture across web, iOS, Android, a Python backend, AI agents and Azure infrastructure, with 4,600+ commits authored since August 2025. Founder of HyeLynk, a community social commerce platform for the Armenian diaspora, and of the Altunyan AI Hedge Fund, an AI-native quantitative research and paper-trading platform. Works across product strategy, system architecture, frontend, backend, AI integration, security and release engineering, with privacy and safety designed in from the start.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, GraphQL

Frontend & mobile: React 19, Next.js 16, React Native (iOS & Android), Expo, Tailwind CSS, UX/UI design

Backend: FastAPI, Fastify, Node.js, Express, REST and GraphQL APIs, WebSockets, microservices

Data: PostgreSQL, Redis, MongoDB, Cloudflare D1, Drizzle ORM, data modeling

AI & ML: LLM applications, multi-agent systems (LangGraph), recommendation engines, health scoring, quantitative research and backtesting, PyTorch, TensorFlow

Cloud & DevOps: Azure (Bicep IaC), AWS, Cloudflare, Docker, Kubernetes, GitHub Actions CI/CD, staged releases

Security: RBAC, MFA, CSRF and rate-limit guards, privacy by design for health data, penetration testing

Product: Product strategy, roadmaps, subscription design, launch governance, brand and packaging direction

PROFESSIONAL EXPERIENCE

Co-Founder & Chief Technology Officer

Aug 2025 – Present

SAECULA (Health Aevum AI Inc.), formerly Healix and AEVUM Health · **saecula.ai** — *Los Angeles, CA*

- Lead product vision, roadmap and technical strategy for an AI-powered health and longevity platform combining assessments, biomarkers, nutrition, fitness, telehealth and personalized supplements.
- Architected a TypeScript/Python monorepo with four client apps (patient web, admin web, iOS, Android), a FastAPI backend, a Fastify API layer and 17 active shared packages; authored 4,600+ commits.
- Designed a traceable recommendation engine that links each assessment answer to health signals, ingredients, domains and a personalized supplement pack, so every recommendation can be explained.
- Directed a multi-agent Health AI Operating System with agents for lab interpretation, nutrition, fitness, supplement safety and clinical workflow support, behind safety controls and human oversight.
- Built a health digital twin and biomarker experience that maps lab results to reference ranges, trends and health domains with data provenance.
- Defined production infrastructure on Azure in Bicep with architecture decision records, machine-enforced architecture rules and a launch scorecard that gates every release.
- Built HIPAA-focused telehealth workflows (scheduling, consent, video visits, clinician tools) behind launch flags; integrated Stripe subscriptions and billing.
- Hardened the platform with edge middleware for rate limiting and CSRF protection, role-based admin access, audit logging and encrypted handling of health data.
- Set up CI quality gates (automated test suites, architecture-rule checks, automated dependency updates) and staged release tracks from restricted web preview and internal mobile beta toward general availability.
- Led two rebrands, Healix → AEVUM Health (Jun 2026) → SAECULA (Oct 2026), across identity, domains, billing, packaging and campaign content.

Founder & Product Architect

2024 – Present

HyeLynk · **hyelynk.ai** — *Los Angeles, CA*

- Designed a social and community commerce platform for the Armenian diaspora with feeds, stories, reels, messaging, a storefront and an admin dashboard.
- Prototyped Instagram-, Threads- and X-style social clients in React Native to define the feed, profile and social-graph experience before committing to the production architecture.
- Architected 16 backend services behind API, GraphQL and WebSocket gateways; built user, content, media, AI orchestration, vision processing and blockchain services in TypeScript and Python.

- Built an AI content moderation pipeline (image and text analysis, safety classification, human review) and a Hedera SDK service for a planned community rewards program.
- Containerized services with Docker Compose and Kubernetes manifests; wrote architecture, security and AI ethics documentation.

Founder & Engineer **2026 – Present**
Altunyan AI Hedge Fund · **altunyanaihedgefund.com** *Los Angeles, CA*

- Built an AI-native research and paper-trading platform (11 Python services, FastAPI, Next.js 15 terminal) with a hard wall between the agents that propose trades and the system allowed to execute them.
- Designed a LangGraph research council (fundamental, technical, sentiment, macro and quant analysts feeding a CIO agent) over point-in-time market, SEC EDGAR, insider, short-interest and news data.
- Wrote a deterministic, property-tested risk engine with final veto, mandatory named human approval, reconcile-before-retry order handling and an append-only, hash-chained ledger and audit log.
- Validate strategies walk-forward at 2× costs against equal-weight and SPY benchmarks and publish the failures; deployed on Azure Container Apps. Paper trading only.

Full-Stack Developer (Client Project) **Aug 2026 – Present**
Philly on the Block · **phillyontheblock.com** *Remote*

- Built and shipped the restaurant's production ordering site, table reservations and staff dashboard with Next.js, React 19, TypeScript and Tailwind CSS 4.
- Integrated Stripe checkout with add-ons, coupons, stock tracking and configurable fees and tax; deployed on Cloudflare with D1 and Drizzle ORM and a one-time setup flow for admin credentials.

Software Engineer & Security Researcher **2023 – 2024**
Independent projects *Los Angeles, CA*

- Built full-stack applications with React, Node.js, Express, PostgreSQL and MongoDB, including GraphQL/Apollo APIs and a Sequelize-style SQL abstraction layer.
- Prototyped an end-to-end encrypted messaging layer with a Signal-compatible key server, self-hosted with Docker.
- Ran penetration tests in a home lab covering SQL injection, XSS, CSRF, SSRF, authentication bypass and privilege escalation; prototyped on-device media moderation with TensorFlow Lite.

SELECTED PROJECTS

Cinematic personal portfolio · *Next.js 16, React 19, Three.js / React Three Fiber, Tailwind 4, GitHub Actions*
Static site with interactive 3D scenes, case studies and typed content files; CI runs type checks, lint and content-accuracy tests before every deploy. Lighthouse 100 SEO and best practices.

SAECULA supplement packaging system · *AI image generation, OCR verification, WebP pipeline*
Directed a 136-pack cover collection built from real capsule photography, with every ingredient label checked against the formula registry by OCR and recorded with image hashes for audit.

AI/ML security framework (research) · *Python, PyTorch*
Explored bias detection, adversarial testing, differential privacy and encrypted inference across the model lifecycle.

UC Berkeley full-stack projects (11) · *JavaScript, Node.js, Express, SQL, MongoDB, React, GraphQL*
Weather dashboard, task board, README generator, SVG logo maker, note taker, employee tracker, MVC tech blog, NoSQL social API, PWA text editor, React portfolio, MERN book search with Apollo GraphQL.

EDUCATION & CONTINUOUS LEARNING

University of California, Berkeley Extension **2024 – 2025**
Full Stack Web Development Program

Coursework: HTML, CSS, JavaScript, Web APIs, Node.js, Express, SQL, ORMs, MVC, NoSQL, PWAs, React, MERN, GraphQL, deployment

Self-directed study in machine learning (PyTorch, TensorFlow), cybersecurity and ethical hacking, distributed systems, cloud architecture and mathematics.

ADDITIONAL

Languages: English (fluent), Armenian (native)

Interests: Health technology, AI safety and evaluation, distributed systems, and building tools for the Armenian diaspora

Creative direction: Naming, wordmarks, packaging systems and AI-assisted campaign visuals and video for SAECULA